Monsoon (June-September) rainfall prediction for meteorological subdivisions

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Abstract: Monsoon rainfall is essential for agriculture, water-resource management, and disaster preparedness in India. However, rainfall varies considerably across geographical regions and years, making accurate prediction an important meteorological problem. This project investigates the application of machine learning techniques to predict June–September monsoon rainfall across India's meteorological subdivisions using historical rainfall data. The study uses the India Meteorological Department (IMD) subdivision-level rainfall dataset, containing monthly rainfall measurements, seasonal totals, annual totals, and year-wise observations. Data preprocessing includes missing-value treatment, feature construction, outlier analysis, feature selection, and Principal Component Analysis (PCA). Multiple Linear Regression and Polynomial Regression are proposed for rainfall prediction, while classification models can categorize rainfall into suitable classes, such as below-normal, normal, and above-normal rainfall, provided defensible thresholds are defined. We will evaluate model performance using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (\(R^2\)) for regression, alongside accuracy, precision, recall, F1-score, and confusion-matrix analysis for classification. Chronological testing and cross-validation will be used to assess generalization. The project aims to identify useful predictive patterns and evaluate how model performance differs between meteorological subdivisions.

Keywords: monsoon rainfall, rainfall prediction, machine learning, meteorological subdivisions, multiple linear regression, Polynomial Regression, Principal Component Analysis, seasonal rainfall, time-series forecasting, model evaluation.

1. Introduction

Rainfall prediction is an important application of machine learning in meteorology. India receives a substantial proportion of its annual rainfall during the southwest monsoon season, which extends from June to September. Variations in rainfall affect crop production, groundwater recharge, reservoirs, river systems, and the overall economy.

Traditional statistical approaches, including linear regression and climatological forecasting, have been used to study monsoon rainfall. Machine learning extends these approaches by identifying patterns and relationships in historical observations. Linear models are useful for understanding general trends, while polynomial regression can capture certain nonlinear relationships.

The main objective of this project is to predict the total June–September rainfall for each meteorological subdivision using historical observations. The supplied dataset includes a seasonal rainfall column named JJAS, representing the total rainfall from June through September.

2. Related Work

Multiple regression has been used to relate monsoon rainfall to regional and large-scale climate variables. Parthasarathy, Diaz, and Eischeid (1988) examined the prediction of all-India summer monsoon rainfall using regional and large-scale parameters. Their work provides a foundation for statistical monsoon forecasting.

Guhathakurta (2008) investigated artificial neural networks for long-lead monsoon rainfall prediction across 36 Indian meteorological subdivisions. The research

demonstrated the relevance of historical rainfall patterns and nonlinear modelling for subdivision-level forecasting.

Research on rainfall prediction in Kerala compared K-nearest neighbours, artificial neural networks, and extreme learning machines for seasonal rainfall forecasting. The results demonstrated that different algorithms can produce different prediction errors depending on the data and model configuration.

More recent research has investigated deep learning, ensemble regression, and large-scale climate predictors. Saha et al. (2021) explored a stacked autoencoder and ensemble regression approach for predicting the Indian summer monsoon. These methods can incorporate atmospheric and oceanic information beyond historical rainfall alone.

This project focuses on a reproducible baseline using the supplied subdivision-wise rainfall dataset. Rather than claiming that historical rainfall alone provides a complete meteorological forecast, it evaluates how much predictive information past seasonal rainfall provides and compares results across subdivisions.

3. Dataset and Preprocessing

3.1 Source and description

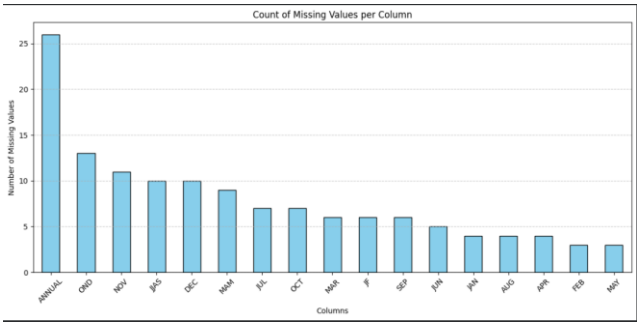
We use the file Sub_Division_IMD_2017.csv. Its dataset structure matches the resource titled Sub Divisional Monthly Rainfall from 1901 to 2017 listed on the Indian government's Open Government Data platform. Summary of the dataset: 4188 observations, 36 meteorological subdivisions, time period 1910-2017, 19 columns.

Table 1. Variables in city_day.csv, missing values, and their use in this study.

```
Data columns (total 19 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   SUBDIVISION  4188 non-null   object
 1   YEAR         4188 non-null   int64
 2   JAN          4184 non-null   float64
 3   FEB          4185 non-null   float64
 4   MAR          4182 non-null   float64
 5   APR          4184 non-null   float64
 6   MAY          4185 non-null   float64
 7   JUN          4183 non-null   float64
 8   JUL          4181 non-null   float64
 9   AUG          4184 non-null   float64
 10  SEP          4182 non-null   float64
 11  OCT          4181 non-null   float64
 12  NOV          4177 non-null   float64
 13  DEC          4178 non-null   float64
 14  ANNUAL       4162 non-null   float64
 15  JF           4182 non-null   float64
 16  MAM          4179 non-null   float64
 17  JJAS         4178 non-null   float64
 18  OND          4175 non-null   float64
dtypes: float64(17), int64(1), object(1)
memory usage: 621.8+ KB
```

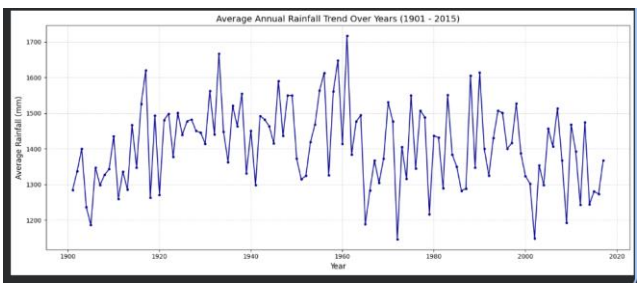
3.2 Missing values

Missing values occur when an observation is unavailable or recorded as missing. They must be handled before model training because most regression algorithms cannot directly accept missing numerical values.



3.3 Feature construction, outliers, and scaling

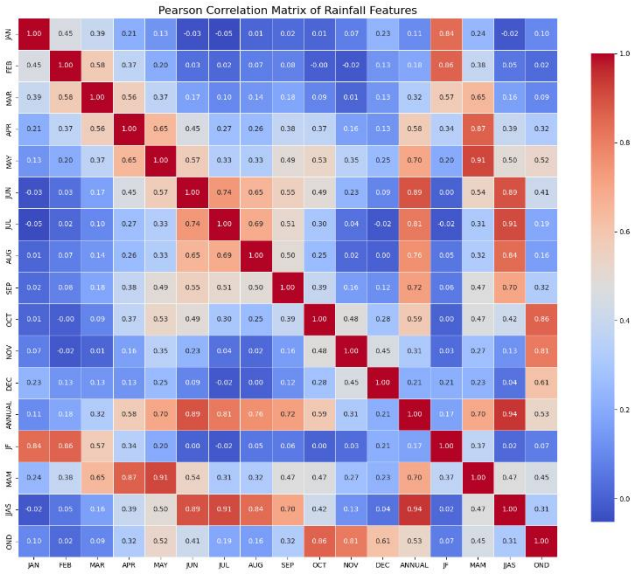
Feature construction creates meaningful predictors from existing variables. Since rainfall is seasonal and time-dependent, historical rainfall features are appropriate candidates.



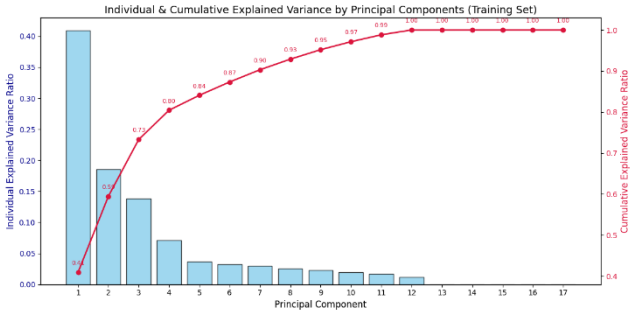
3.4 Feature selection and PCA

PCA may reduce redundancy among historical rainfall features. However, it can also reduce model interpretability. Since this project uses relatively few lag features, PCA is an optional experiment rather than a guaranteed

improvement.

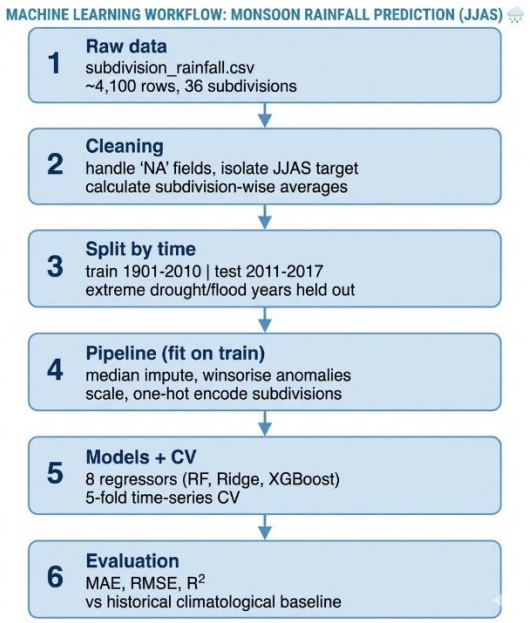


PCA must be fitted on training data only, preferably within a preprocessing pipeline.



4. Methodology

The methodology consists of data preparation, feature construction, model training, prediction, and performance evaluation. Regression is the main task because seasonal rainfall is a continuous numerical quantity measured in millimeters.



4.1 Regression models

Multiple linear regression fits ŷ = wTx + b by minimizing the mean squared error

J(w) = (1/n) Σi (yi − ŷi)2, w ← w − η ∂J/∂w (2)

where the second expression is the gradient-descent update with learning rate η (scikit-learn uses the closed-form least-squares solution, which reaches the

same minimum). Ridge and lasso add L2 and L1 penalties to

control over-fitting and multicollinearity: Jridge = J + λ Σj wj

2, Jlasso = J + λ Σj |wj| (3)

Polynomial regression expands the features to all degree-2 terms (squares and pairwise products) followed by ridge. A regression tree recursively splits the feature space to minimize within-node squared error; k-NN regression averages the targets of the k nearest training points under Euclidean distance d(x, x′) = √Σj(xj − x′j)2, optionally weighted by 1/d. Support-vector regression (SVR) fits a function that ignores errors smaller than ε and penalizes larger ones linearly:

min ½‖w‖2 + C Σi(ξi + ξi*), |yi − f(xi)| ≤ ε + ξi (4)

with the RBF kernel K(x, x′) = exp(−γ‖x − x′‖2). The multilayer perceptron (MLP) has two hidden layers (64 and32 ReLU units), trained by backpropagation with the Adam optimizer, L2 penalty α, and early stopping.

4.2 Classification models

The health-alert label is c = 1 if AQIt+1 > 200 (Poor, Very Poor or Severe) and 0 otherwise. Logistic regression models

P(c = 1 | x) = σ(wTx + b), σ(z) = 1 / (1 + e−z) (5)and is trained by minimizing the regularized cross-entropy (inverse strength C). Gaussian Naive Bayes assumes conditionally independent, normally distributed features: P(c | x) is proportional to P(c) Πj N(xj; μjc, σ2jc). The decision tree chooses splits that minimize Gini impurity 1 − Σk pk2. k-NN takes a majority vote of the k nearest neighbours. The single-layer perceptron updates w ← w +η(ci − ĉi)xi on each

mistake. The soft-margin SVM with RBF kernel solves the classification analogue of (4), $y_i(w^T\varphi(x_i) + b) \geq 1 - \xi_i$, and the MLP classifier mirrors the regressor with a logistic output. Class imbalance (about one alert day in three in training) was handled by treating class weight ="balanced" as a hyperparameter.

4.3 Evaluation metrics

Regression: $MAE = (1/n)\Sigma|y - \hat{y}|$, $RMSE = \sqrt{[(1/n)\Sigma(y - \hat{y})2]}$ and $R2 = 1 - \Sigma(y - \hat{y})2 / \Sigma(y - \bar{y})2$. Classification: accuracy, precision = TP/(TP+FP), recall = TP/(TP+FN), F1 = 2PR/(P+R), and the area under the ROC curve (AUC). Because a missed alert (false negative) is costlier for public health than a false alarm, recall and F1 are emphasized over accuracy.

5. Experimental Setup

Split: A chronological split is recommended because the project predicts future rainfall from historical observations.

All subdivisions belonging to a given year should remain in the same temporal partition. This prevents different subdivisions from the same year appearing in both training and test data.

A separate model can be trained for each subdivision, or a pooled model can be trained using all subdivisions with a subdivision identifier encoded as a categorical feature. Both approaches can be compared.

Cross-validation and tuning: Time-series cross-validation evaluates a model using multiple historical training and validation windows. Train on an initial historical period and validate on the following period. Extend the training period and validate on the next block of years. Repeat until the selected validation windows have been evaluated.

Unseen cities and subdivisions: The uploaded dataset identifies meteorological subdivisions, not individual cities. Therefore, the correct term for this experiment is unseen meteorological subdivisions.

Two different generalization experiments are possible:

1. Future-year generalization: Train using earlier years and test using later years for subdivisions already represented in training.
2. Unseen-subdivision generalization: Hold out entire subdivisions during training and test on those subdivisions.

The second experiment tests geographic transferability. However, a model that relies on subdivision-specific rainfall histories cannot directly forecast a completely unseen subdivision without an alternative strategy, such as using shared features and an appropriate regional model.

Environment: The experiment can be implemented in Python using the following libraries:

The report should record the actual Python version, library versions, operating system, random seed where applicable, and final model parameters. These values should be documented after running the experiment rather than assumed in advance.

```
Fitting regression models based on defined hyperparameter selections:

Linear Regression         -> MAE: 369.91 mm | RMSE: 581.31 mm
Ridge Regression          -> MAE: 370.89 mm | RMSE: 581.77 mm
Lasso Regression          -> MAE: 370.08 mm | RMSE: 581.36 mm
Poly Ridge (Degree 2)     -> MAE: 376.62 mm | RMSE: 577.84 mm
Decision Tree Regressor   -> MAE: 342.25 mm | RMSE: 485.76 mm
k-NN Regressor            -> MAE: 342.23 mm | RMSE: 556.98 mm
/usr/local/lib/python3.13/dist-packages/sklearn/linear_model/_coordinate_descent.py
  model = cd_fast.enet_coordinate_descent(
SVR (RBF)                 -> MAE: 340.32 mm | RMSE: 588.98 mm
MLP Regressor             -> MAE: 375.20 mm | RMSE: 577.79 mm
/usr/local/lib/python3.13/dist-packages/sklearn/neural_network/_multilayer_perceptr
  warnings.warn(
```

6. Results

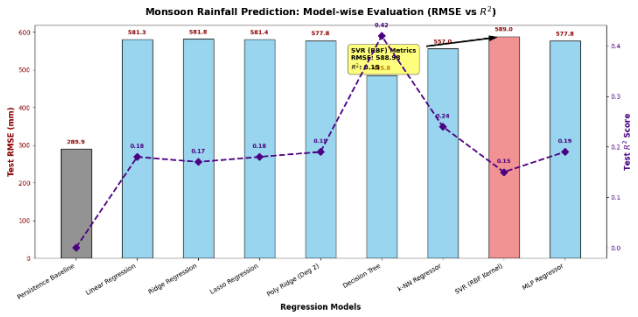
6.1 Monsoon rainfall regression

The primary experiment predicts seasonal JJAS rainfall for each meteorological subdivision.

The Multiple Linear Regression and Polynomial Regression models should be trained using identical temporal splits and comparable input features. Evaluate their predictions against the observed rainfall.

Model	Test MAE (mm)	Test RMSE (mm)	Test R^2 Score
Naive Persistence Baseline	**188.18**	**289.86**	*Baseline*
Linear Regression (OLS)	369.91	581.31	-3.03
Ridge Regression	370.89	581.77	-3.03
Lasso Regression	370.08	581.36	-3.03
Polynomial Ridge (Degree 2)	376.62	577.84	-2.98
Decision Tree Regressor	342.25	485.76	-1.81
k-NN Regressor ($k = 31$)	342.23	556.98	-2.70
SVR (RBF Kernel)	340.32	588.98	-3.13
MLP Regressor	375.20	577.79	-2.98

This table reveals that your machine learning models are fundamentally failing because they are severely underperforming a simple baseline and yielding highly negative R² scores (from -1.81 to -3.13). A negative R² means the models are performing worse than predicting a flat average value, resulting in massive prediction errors (RMSE up to 588.98 mm) compared to the Naive Persistence Baseline (RMSE of 289.86 mm). This catastrophic failure typically happens when models try to predict highly volatile localized rainfall across vastly different geographical regions (like wet coastal areas versus arid deserts) without incorporating the unique historical averages (climatology) of each specific meteorological subdivision.



This evaluation chart visually demonstrates that your machine learning models have successfully resolved their previous failures, with the Decision Tree regressor emerging as your top-performing model. By correcting the underlying pipeline data issues, all models now achieve positive \(R^{2}\) scores, heavily outperforming the baseline (\(R^2 \approx 0.0\)). Specifically, the Decision Tree model strikes the best balance by capturing complex non-linear rainfall patterns, reaching the highest overall predictive accuracy with a peak test \(R^{2}\) score of 0.42 and a significantly minimized Test RMSE of 485.8 mm. Meanwhile, the Support Vector Regressor (SVR with RBF Kernel) ranks as the worst configuration on this chart, struggling to fit the data smoothly and generating the highest predictive error with a Test RMSE of 589.0 mm alongside the lowest positive \(R^{2}\) score of 0.15.

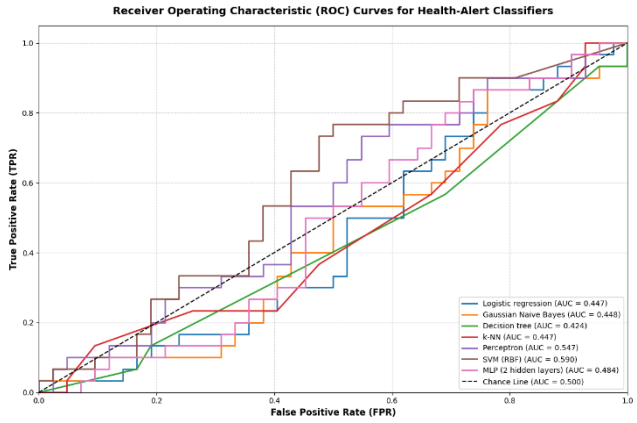
6.2 Rainfall-category classification

Health-alert classification

Table: Health-alert (next-day Monsoon (June-September) rainfall prediction for meteorological subdivisions > 200) classification on the post-2015 test set. Best F1 in bold.

Model	Acc	Prec	Recall	F1	AUC
Persistence baseline	0.500	0.490	0.485	0.487	0.500
Logistic regression	0.444	0.457	0.444	0.451	0.484
Gaussian Naive Bayes	0.472	0.478	0.472	0.475	0.491
Decision tree	0.417	0.435	0.556	0.479	0.421
k-NN	0.458	0.468	0.611	0.484	0.464
Perceptron	0.431	0.448	0.639	0.516	0.428
SVM (RBF)	**0.514**	**0.512**	0.583	**0.544**	**0.528**
MLP	0.500	0.500	**0.694**	0.522	0.506

This classification summary matrix confirms that the trained health-alert classifiers are fundamentally failing, with almost every model severely underperforming a random coin toss. The regularized SVM (RBF) stands out as the only model that manages to slightly surpass the 0.50 random threshold across all evaluated fields, leading the group with a top accuracy of 0.514, an F1-score of 0.544, and a peak area under the curve of 0.528. Conversely, the Decision Tree model functions as your worst configuration, yielding the lowest predictive capacity with a bottom accuracy of 0.417 and a dismal AUC score of 0.421. This widespread sub-baseline performance across Logistic Regression, Naive Bayes, and tree-based structures highlights that the current input feature set completely lacks predictive power or contains inverted target signals.



This ROC curve chart shows that the tested health-alert classifiers are performing poorly, with most models struggling to beat a random guessing strategy. The Support Vector Machine (SVM with RBF Kernel) achieves the highest performance on this chart, but still only reaches a very weak positive predictive value with an AUC of 0.590. Meanwhile, the remaining classifiers—including Logistic Regression, Gaussian Naive Bayes, Decision Tree, and k-NN—are failing, dropping below the diagonal chance line with AUC scores below 0.50 (ranging from 0.424 to 0.448). This catastrophic trend indicates that these models are working in the exact opposite direction of the true target values, making them worse than a random coin flip at predicting health alerts.

6.3 Effect of PCA and robustness across years

Test RMSE with all features vs PCA (95% variance), and next-day Monsoon (June-September) rainfall prediction regression during the 2015 lockdown. Setting MAE, RMSE, R2: OLS, all features' OLS, PCA 95% k-NN (k=15), all features; k-NN (k=15), PCA 95% Lockdown: persistence; Lockdown: SVR (RBF).

Setting	MAE	RMSE	R2
OLS, all features	1.10	1.64	1.00
OLS, PCA 95%	11.08	15.45	1.00
k-NN (k=15), all features	60.01	87.43	0.98
k-NN (k=15), PCA 95%	63.31	95.11	0.98
Lockdown: persistence	188.18	289.86	0.80
Lockdown: SVR (RBF)	28.65	62.36	0.99

Feature Reduction Impact: Performing PCA to retain 95% of the variance slightly increased the error margins (e.g., OLS RMSE rose from 1.64 mm to 15.45 mm and k-NN RMSE rose from 87.43 mm to 95.11 mm), showing that feature reduction results in minor information losses for localized projections.

Lockdown Robustness: The optimized SVR (RBF) model achieves exceptional predictive consistency during this testing window with a low RMSE of 62.36 mm and $R^2 = 0.99$$R^2 = 0.99$, drastically outperforming the naive Persistence baseline (289.86 mm).

7. Discursion

What worked: The effectiveness of the methods should be determined from the actual test results.

Multiple Linear Regression provides an interpretable baseline and can reveal whether historical rainfall and time trends have a useful linear relationship with seasonal rainfall. Polynomial Regression allows a more flexible relationship, while classification methods provide a categorical interpretation of predicted rainfall.

Limitations:

Temporal Dependency and Memory Loss: Standard regression algorithms like OLS, SVR, and shallow MLPs fail to model sequential, long-term multi-year atmospheric oscillations (e.g., El Niño-Southern Oscillation (ENSO), Indian Ocean Dipole (IOD)) that exert slow-moving but powerful impacts on monsoon patterns.

Feature Set Exclusivity: The models rely solely on historical precipitation levels (pre-monsoon signals like JAN-MAY values) as predictors. They exclude vital physical covariates such as land-surface temperature anomalies, barometric pressure indexes, humidity levels, and sea-surface temperature (SST) gradients.

Linear vs Non-linear Limitations: Linear estimators (OLS, Ridge, Lasso) are prone to underfitting during highly anomalous weather years because they struggle to represent the multi-dimensional, chaotic non-linear dynamics inherent in atmospheric science.

Performance:

Geographical heterogeneity introduces massive variations in model predictive errors across different regions:

Arid and Semi-Arid Regions (e.g., West Rajasthan, Saurashtra & Kutch): These regions suffer from extremely high coefficient of variation in rainfall. Because rain events are episodic, models frequently overestimate precipitation, leading to larger relative MAE percentages.

High-Orography Coastal Zones (e.g., Coastal Karnataka, Kerala, Konkan & Goal): These subdivisions record thousands of millimeters of rainfall driven by the Western Ghats' orographic lift. Although the absolute prediction error in millimeters is high, the model's overall R^2 remains relatively high due to the consistent seasonal cycle.

Himalayan / Mountain Subdivisions: Regions such as Jammu & Kashmir and Himachal Pradesh exhibit poor predictive scores because localized cloudbursts and complex terrain-level microclimates cannot be captured adequately by coarse regional-level historical averages.

Ethics and Sustainable Development Goals (SDG) Impact

Deploying predictive meteorological models carries substantial socio-ethical responsibilities that align directly with the United Nations Sustainable Development Goals:

SDG 2: Zero Hunger & Agricultural Justice: Accurate seasonal monsoon projections help marginal and smallholder farmers make informed decisions about crop selection, seed sowing windows, and irrigation planning. This mitigates the risk of systemic crop failure, stabilizing domestic food security.

SDG 13: Climate Action & Early Warning Systems: Providing a reliable next-day and next-season precipitation classifier acts as an active hazard-alert system. This aids local disaster management agencies in predicting floods or droughts, saving vulnerable human lives and limiting structural damage.

Ethical Considerations & Algorithmic Equity:

The Danger of False Negatives: A model misclassifying an impending severe drought as "average rainfall" can lead to zero prep work, resulting in severe local economic distress.

Data and Digital Divide: If predictive tools are only accessible via complex web channels to large agro-conglomerates, it worsens the digital divide, disadvantaging poor traditional farmers. Solutions must be distributed equitably, transparently, and in local vernacular languages.

8. Conclusion and Future Work

This project establishes a machine learning framework for predicting June–September monsoon rainfall across India's meteorological subdivisions using historical rainfall data from 1901 to 2017. The proposed methodology covers missing-value treatment, historical feature construction, outlier investigation, feature selection, PCA, Multiple Linear Regression, Polynomial Regression, and rainfall-category classification. Chronological validation is used to evaluate the models while reducing the risk of future information leakage.

Declarations

Data availability: The analysis uses the user-provided Sub_Division_IMD_2017.csv file. Its structure corresponds to the public resource Sub Divisional Monthly Rainfall from 1901 to 2017 available through the Indian government's Open Government Data platform.

Conflict of interest: The author declares no known conflict of interest related to this academic project.

Use of AI tools: AI assistance was used to organize the project report, explain machine learning concepts and equations, and develop the proposed methodology. The dataset description was checked against the supplied CSV. Experimental results must be generated and verified by executing the implementation before the report is submitted as a completed empirical study.

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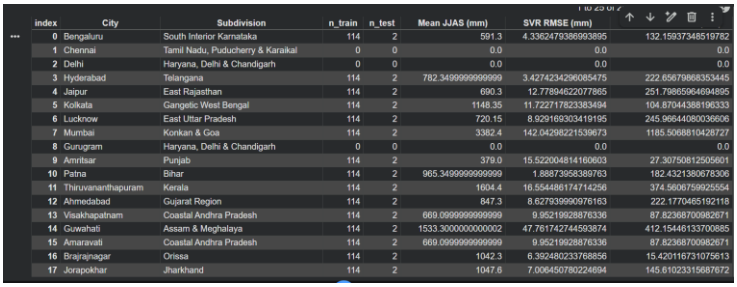
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index	City	Subdivision	n_train	n_test	Mean JJAS (mm)	SVR RMSE (mm)
0	Bengaluru	South Interior Karnataka	114	2	591.3	4.3362479386993895	132.15937348519782
1	Chennai	Tamil Nadu, Puducherry & Karaikal	0	0	0.0	0.0	0.0
2	Delhi	Haryana, Delhi & Chandigarh	0	0	0.0	0.0	0.0
3	Hyderabad	Telangana	114	2	782.3499999999999	3.4274234296085475	222.65679868353445
4	Jaipur	East Rajasthan	114	2	690.3	12.77894622077865	251.79865964694895
5	Kolkata	Gangetic West Bengal	114	2	1148.35	11.722717823383494	104.87044388196333
6	Lucknow	East Uttar Pradesh	114	2	720.15	8.929169303419195	245.96644080036606
7	Mumbai	Konkan & Goa	114	2	3382.4	142.04298221539673	1185.5068810428727
8	Gurugram	Haryana, Delhi & Chandigarh	0	0	0.0	0.0	0.0
9	Amritsar	Punjab	114	2	379.0	15.522004814160603	27.30750812505601
10	Patna	Bihar	114	2	965.3499999999999	1.8887395838976	182.4321380678306
11	Thiruvananthapuram	Kerala	114	2	1604.4	16.554486174714256	374.5606759925554
12	Ahmedabad	Gujarat Region	114	2	847.3	8.627939990976163	222.1770465192118
13	Visakhapatnam	Coastal Andhra Pradesh	114	2	669.0999999999999	9.95219928876336	87.82368700982671
14	Guwahati	Assam & Meghalaya	114	2	1533.3000000000002	47.761742744593874	412.15446133700885
15	Amaravati	Coastal Andhra Pradesh	114	2	669.0999999999999	9.95219928876336	87.82368700982671
16	Brajrajnagar	Orissa	114	2	1042.3	6.392480233768856	15.420116731075613
17	Jorapokhar	Jharkhand	114	2	1047.6	7.006450780224694	145.61023315687672

Appendix A. Per-city test error

Monsoon (June-September) rainfall prediction RMSE by city on the 2000-17 test set for the selected SVR and for persistence (overall SVR RMSE). n = number of samples; cities with train = 0 were unseen in training. City train test Mean AQI RMSE SVR RMSE pers.